

Men’s basketball assortativity — why empirical sorting looks near zero

Prepared: September 25, 2026
Author lane: COMPASS (scientific framing; not a pipeline audit)
Trigger: Charles debrief after long conversation with Alex — pause assortativity sensitivity grid until pool composition and performance metric are re-examined.
Standalone: This memo states the scientific worry, separates “real world” sorting from measured **H_sort**, lists confounders tied to points-per-minute and roster construction, and proposes a ordered diagnostic menu. It does **not** authorize new runs.
Same folder (read separately — not merged here): [SCOUT evidence note](#) · [VECTOR scientific note](#). Section **11** below adds COMPASS-only reactions after reading those two.

1. The core intuition (and why it is reasonable)

Common-sense claim: Division I men’s basketball is not a league of random talent placement. A roster like Duke or UConn is not a flat mix of mediocre and elite players in the way a naïve “no sorting” story would suggest. Most players who **earn meaningful minutes** on those teams are **very good** relative to the national pool.
Empirical readout today: On the locked hero panel (minimum twenty season minutes, eleven captured games per team-season, within-season points-per-minute z-score, leave-one-out teammate quality), realized sorting index **H_sort** is **small** (bracket calibrations often find homophily parameter $\rho \approx 0$ matches empirical **H_sort**). That can sound like “basketball has no assortativity,” which conflicts with common sense.
COMPASS read: The conflict is usually **not** “NCAA is actually random.” It is “**our measured partition + ability coordinate + peer definition compress sorting into a statistic that looks like zero.**” That is a **measurement and pool-definition** problem before it is a “reject homophily in the model” problem.

2. Two boundary conditions that dominate the basketball story

Alex and Charles named two scarce-resource limits that Army does not share in the same form:

Boundary	Basketball	Why it matters for us
Scarce selection	Draft rate $K/N \approx 1\%$	Hero curve is estimated on a tiny positive class; tail behavior is noisy; small denominator moves look like “no assortativity effect.”
Scarce on-court slots	Five players at a time	Congestion for “who gets seen / who gets the ball / who gets the stat line” is about rotation and minutes , not “fifteen names on a roster sheet.”

The second point is the new emphasis from today’s conversation. Our generative story speaks about **peer pools** and **congestion in the score**, but much of the empirical panel still defines peer context from **season-long roster aggregates** (leave-one-out mean teammate points-per-minute z-score over everyone who clears season filters).
Mismatch in one sentence: We are modeling **congestion as if the relevant peer group is the full listed roster**, while the **sport’s scarce interaction slot is five players on the floor**.
That mismatch does not prove assortativity is high; it explains why **congestion measured through full-roster, rate-based points per minute** may **understate** the strategic peer pressure draft-relevant players actually felt.

3. How points-per-minute helps — and how it confounds sorting and congestion

3.1 Why points per minute was attractive

Total season points confounds **talent** with **opportunity** (seniority, rotation trust, injury, blowouts). Points per minute asks: “**What did you produce per unit of playing time?**” That aligns with “talent should not be proxied by minutes alone.”

3.2 How points per minute interacts with sorting index H_sort

Ability on the main panel is **within-season z-scored points per minute** across **all retained player-seasons** in that year (after filters). That construction:

- Forces a single national spread** each season — everyone is expressed relative to the same cross-section.
- Preserves substantial individual dispersion** — histograms of $\hat{A}_i$ are wide, not collapsed.
- Does not guarantee large between-team dispersion** in team means T_j or in leave-one-out pool quality — those depend on **who is averaged together** on each roster.

Charles’s “huge bands of talent at all levels” is **true at the player level** in z-space. Assortativity in our index is **not** “are players different?” It is “**do team labels explain much of that player-level variance?**” A league can have wide player dispersion and still show **low H_sort** if rosters are **similarly wide** team-to-team (each team has stars, rotation, and bench noise in comparable proportions).
Elite programs in z-space: A “good” player at a mid-major and a “good” player at a blueblood can sit **not far apart** in within-season z after filtering to rotation players — both are above average producers in their contexts, but z-scoring **pools contexts for interpretation and hero bins**. Team means move, but not necessarily enough to push **H_sort** toward Army-like values without stronger peer definitions.
Correction (after VECTOR / SCOUT): Within a season, **within-season z-scoring is an affine transform** of raw points per minute — it **does not by itself raise or lower H_sort** (numerator and denominator of the sorting index scale together). What moves **H_sort** is **who is retained, raw rate construction, and team assignment**, not the z step alone. Earlier prose here that sounded like “z hides tier → low H_sort” was **imprecise**; z matters for **comparability language and leave-one-out hero**, not for the sorting-index algebra.

3.3 How points per minute torpedoed the congestion narrative (mechanism, not data bug)

Congestion in the score is meant to capture **crowding among viable peers** for a scarce outcome (draft attention, role, touches). Points per minute:

- Rewards efficiency in low minutes** — a player who scores in a short stint can look **deceptively strong** per minute.
- Is extremely noisy at low minutes** — denominators make rates unstable; zero-point games create literal zeros.
- Is partly an outcome of role and lineup** — who you share the floor with affects shots, not just “who you are.”

So points per minute is doing **double duty**: it is both **ability input** and a **shadow of on-court opportunity structure**. When bench players with **few minutes per game** remain in the **peer denominator**, they inflate roster size without representing **on-court competition** for the five simultaneous slots.
Charles’s instinct — “**I don’t trust points per minute for guys who don’t play more than five minutes in a game**” — targets exactly that: **wrong peers + wrong signal**, not merely “outliers.”

4. What the hero tail still shows (and what it does not)

On the **post-box-quality-control** panel, the **elite right-tail dip** in draft rate is **much weaker** than Army’s, and under some specs it **disappears** as a concave quadratic (flat top ventiles). That pattern fed the worry: “**Maybe assortativity does not matter in the model.**”

Separate two claims:

Claim	Evidence status
Peer context matters for draft odds (middle rise in hero bins)	Still supported on defended panel
Elite-bin downturn is robust	Not supported on post-QC leave-one-out axis — much of July dip was contamination
Low H_sort means no real sorting in nature	Does not follow — index is partition- and metric-dependent
Low calibrated p means Levine–Gates cannot sort	Does not follow — bracket matches realized H_sort, not “true homophily in the world”

The hero and H_sort answer different questions. You can have a hero shape that responds to leave-one-out pool quality while H_sort on the same abilities and rosters stays small because team labels explain little variance in the chosen coordinate.

5. Why we might not “see” assortativity empirically — candidate explanations

Ordered from most aligned with today’s conversation to worth testing but secondary.

A. Peer pool definition (full roster vs rotation vs on-court unit)

- **Current default:** Leave-one-out mean teammate perf over roster members on (team_id, season) after season filters.
- **Structural issue:** Fifteen-man roster ≠ five on-court congestion; walk-on and deep-bench minutes dilute or distort means.
- **Charles direction:** Exclude or down-weight players below five minutes per game (or similar) from peer context, not necessarily from the focal sample — mirror Army “drop zero–top-block peers from pool mean.”

B. Performance metric (points per minute and its z-score)

- Rate metric + short minutes → unstable $\hat{A}_i$ and polluted teammate means.
- Alternatives (box plus, offensive rating, minutes as separate channel) were explored; points per minute won the hero shape gate for the old peer definition — not necessarily for a rotation-restricted pool.

C. Individual season minutes floor (twenty total minutes)

- **Production lock:** drop player-seasons with < 20 total season minutes.
- **Effect:** Removes many non-rotation rows from both focal set and peers (when rebuild path is used).
- **Does not fix:** Per-game floor; stars’ peers can still include low-minute-per-game bench players who cleared twenty season minutes.

D. Team captured-games filter (eleven games)

- Removes fragmentary one-game opponents; does not enforce Division I membership or full schedule completeness.
- **Weak teams with partial schedules** were part of the τ_j inflation** worry in the VECTOR history review.

E. Population scope (all teams vs Division I vs draft-producing programs)

Restriction	Intent	Risk
All retained teams (ALLT hero)	General NCAA estimand	Mixes tiers; z-scores absorb tier in one national scale
Division I only	Cleaner comparability	Needs defensible membership rule in extract
Teams with ever-drafted player (+DFT)	“Places that produce pros”	Selection on outcome; assortativity and hero both shift — fine as sensitivity, dangerous as only estimand
Power conferences / tiers	Explicit stratification	Smaller N; different scientific claim

COMPASS view: “Draft-ever teams only” is a reasonable sensitivity for the Squid-vs-Jackals story (comparable ambition environments), not a substitute for fixing peer definition on the main panel.

F. Within-season standardization itself

- Z-scoring within season changes units and verbal comparability, not H_sort on a fixed retained sample (see §3.2 correction).
- Common-sense “everyone on Duke is good” is about latent talent / recruiting tier; points-per-minute z is observed scoring rate — defenders and role players need not rank “elite” on that scalar even when they are elite basketball players.

G. Leave-one-out vs team mean axis

- Hero bins on poolq_loo; τ_j ** diagnostics can look more separated than leave-one-out suggests.
- Sorting index can differ if computed on partitions that emphasize team means vs teammate exclusion.

6. Direct answers to Charles’s “what are we missing?” list

Question	Short answer
Should there be talent assortativity in men’s Division I basketball?	Yes in the real institution — recruiting, scholarships, and role allocation sort talent. Not necessarily high in our H_sort on this panel.
Is it the performance metric?	Partly. Points per minute conflates rate talent with role and low-minute noise; it fights the congestion story when peers include non-competitors.
Is it the twenty-minute season filter?	Partly. It helps but is coarse; it does not encode five on court or per-game rotation.
Is it the team games filter?	Partly for fragmentary opponents, not for the Duke-vs-mid-major logic.
Division I only?	Worth a labeled sensitivity if membership is definable in data.
Draft-ever teams only?	Worth a sensitivity, not the sole fix; conditions on the outcome ecosystem.
Are we missing assortativity, or mis-measuring pools?	Prioritize mis-measured pools and peer sets before concluding $p = 0$ is “truth.”

7. Proposed “one more thing” before assortativity sensitivity grid

This aligns with what Charles told Alex — re-look pool composition driven by metric + filters, not another p sweep on the same panel.

7.1 Scientific hypotheses to test (document only)

1. **Rotation peer hypothesis (hero / congestion axis):** Leave-one-out pool quality recomputed over **rotation-eligible teammates only** can move the empirical **draft curve** and change **interpreted congestion** even when focal $\hat{A}_i$ is fixed — **without changing H_{sort}** (VECTOR distinction).
2. **Sample / metric hypothesis (H_{sort}):** H_{sort} and bracket p^* change only if the **retained athlete set**, **raw points-per-minute definition**, or **team labels** change — e.g. drop unstable low-minute players from the panel, or redefine rate with a minutes floor in the denominator. Direction **not guaranteed** to rise.
3. **On-court congestion hypothesis (generative):** Team L_C in the Levine–Gates score uses **full eligible roster** in planned assort runs; a rotation-restricted empirical peer average is **not** interchangeable without declaring a **different generative model** (VECTOR construction spec).
4. **Rate vs level hypothesis:** A second performance construct (not a silent swap for points per minute) may show higher between-team dispersion — each measure needs its own missingness audit.
5. **Tier stratification hypothesis:** H_{sort} may read higher **within** conference or draft-producing subsets — separate estimand, not a fix for the national panel.
6. **Null benchmark hypothesis:** $H_{\text{sort}} \approx 0.06$ is not self-interpreting; compare to **exchangeable random team assignment** expectation $(J-1)/(N-1)$ on the **same** N , J , and ability vector (VECTOR; not yet run on the freeze population).

7.2 Ordered diagnostic menu (for SCOUT / VECTOR when Charles authorizes runs)

VECTOR’s bounded first step (COMPASS agrees this should precede a long menu): one **2015 rotation audit** — minutes, appearances with positive minutes, minutes per appearance; share of peer pool below a provisional five-minute-per-game rule; **same focal players**, leave-one-out with vs without low-minute teammates in the peer mean only. **Gate:** settle **initial execution stop** (points-with-missing-minutes, canonical team) before calling results final.

Broader menu (after audit, if warranted):

1. **Describe** peer dilution — define **denominator** for any “per game” rule (games with positive minutes vs all captured games).
2. **Recompute leave-one-out only** under peer exclusions — **hold focal $\hat{A}_i$ and team assignment fixed**; report hero ventiles.
3. **If sample or PPM definition changes:** recompute H_{sort} , permutation null, and p bracket on the **new** population — do not expect step 2 alone to move H_{sort} .
4. **Optional labeled subsets:** season-specific Division I membership crosswalk; draft-ever teams — sensitivity only.
5. **Do not** subset or threshold-hunt to **recover** a preferred inverted-U (VECTOR warning; COMPASS agrees).

7.3 What not to do

- Do **not** treat $p \approx 0$ on the current panel as proof that homophily is absent in college basketball.
- Do **not** drop assortativity from the model **until** peer pool and metric sensitivity is documented.
- Do **not** conflate fixing the hero tail with restoring **low** elite-bin draft rates from pre-QC contamination.

8. Squid team vs Jackals team (framing for the paper)

Charles’s Squid / Jackals metaphor is **about tier and opportunity**, not literal team names:

- **Same human capital type** (good-to-elite player) can face **different congestion** on a **contending roster** vs a **mid-tier roster**.
- Points-per-minute z-scores **partially equate** those players across institutions, which helps comparability but **blurs** the institutional sorting story H_{sort} is supposed to detect.
- The empirical hero asks about **leave-one-out teammate quality vs draft**, which **does** speak to Squid-vs-Jackals **conditional on making the panel** — but only if **teammates in the average are the ones who actually shared rotation and opportunity**.

One sentence that summarizes the research pivot: *We are not abandoning assortativity; we are testing whether near-zero H_{sort} is an artifact of full-roster, rate-based peers in a five-on-court sport.*

9. Related repo documents

Topic	Path
POST-QC hero vs pre-QC tail	3-Master_Plan/re_entry/HEROs_and_PASSes/PD20_22_campaign_big_picture.md
Data hygiene history (SCOUT)	3-Master_Plan/VECTOR_work/new_VECTOR_work/assort_analysis/docs/source_review/ASSORT_20260925_SCOUT_response_data_hygiene_and_model_history.md
Campaign framing (COMPASS)	.../ASSORT_20260925_COMPASS_addendum_data_hygiene_and_model_history.md
Assign / score / select reference	3-Master_Plan/re_entry/LG_model_desk_reference.md
Reigning hero lock spec	3-Master_Plan/re_entry/HEROs_and_PASSes/sports_sandbox/reigning_hero/README.md
Binding: score ≠ select	3-Master_Plan/BINDING_Selection_is_its_own_step.md

10. Suggested next step for Charles

1. **Socialize** this framing with Alex — confirm the **one-more-thing** is **peer-pool / per-game minutes**, not another global p sweep.
2. **Authorize SCOUT** (when ready) to implement **leave-one-out recomputation** with **peer eligibility rules** as a **sensitivity branch**, with provenance JSON mirroring Pass A sensitivity discipline.
3. **Freeze** assortativity investigation population **after** that branch, not before.

11. Additions after reading SCOUT and VECTOR (same day)

These are COMPASS-only reactions — not a merged consensus document.

11.1 Separate estimands (borrowed clarity from SCOUT)

Object	One-line question
Empirical H_sort	Do team labels explain much of $\hat{\mathbf{A}}_i$ variance on real rosters?
Calibrated $\mathbf{p}^*$	What assign homophily matches that partition? (On post-QC panel, $\mathbf{p}^*$ often hits the $\mathbf{p}=0$ boundary because empirical H_sort can sit below the sim mean at $\mathbf{p}=0$ — nearest boundary, not “exact random match.”)
Assort experiment	Does $\mathbf{p}$ or λ move top-K winners at fixed roster?
Hero	Draft rate vs poolq_loo — outcome layer; not H_sort

Low $\mathbf{p}^*$ is not the same claim as “basketball has no recruiting sort in the real world.”

11.2 Five on court — refine, do not caricature (VECTOR)

- More than five players rotate over a season; position and role split competition for minutes and shots.
- Teammates on court can help scoring, not only crowd it — congestion is not “count bodies $\geq$ five.”
- Low minutes per appearance may reflect selection (depth burying a good player on a great team), not automatic “non-competitor.” COMPASS still favors reliability and peer-definition probes; VECTOR warns against moralizing the cutoff.

11.3 Saved magnitude anchor (SCOUT)

Post–box–QC, H_sort $\sim 0.06\text{--}0.07$ on defended panels is weak sorting, not literally zero. Pre–QC 2015 ≈ 0.114 in an Aug 14 bracket file is not the same population as today’s ≥ 11 games panel (≈ 0.061 in Aug 19 trace). Do not mix when telling the story.

11.4 Points per minute $\neq$ latent “Duke is all good” (VECTOR)

Elite programs may recruit stronger latent talent while observed scoring rate stays a narrow, role-sensitive outcome. Small H_sort on points-per-minute z does not falsify Charles’s institutional intuition — it bounds what that measure says about between-team dispersion.

11.5 Where COMPASS still pushes beyond VECTOR’s minimal audit

- Align generative congestion with sport structure remains a model-design thread for after the descriptive audit — not something the hero-only leave-one-out tweak resolves.
- Army-style peer exclusion (drop non-competitors from pool mean only) is still a useful analogy for rotation-restricted leave-one-out, even though it does not touch H_sort by itself.